

Problem 12

Evaluate the integral:

$$\int 5^x 3^{-x} dx.$$

Solution

To solve the integral $\int 5^x 3^{-x} dx$, we start by rewriting the integrand for simplification.

Step 1: Simplify the integrand

The integrand is $5^x 3^{-x}$. Using the property of exponents, $a^m b^n = (a/b)^m$ when $m = n$, we can combine the terms:

$$5^x 3^{-x} = \left(\frac{5}{3}\right)^x.$$

Thus, the integral becomes:

$$\int 5^x 3^{-x} dx = \int \left(\frac{5}{3}\right)^x dx.$$

Step 2: General formula for exponential integrals

The integral of a^x with respect to x is:

$$\int a^x dx = \frac{a^x}{\ln(a)} + C, \quad \text{for } a > 0, a \neq 1.$$

Here, $a = \frac{5}{3}$. Applying this formula, we get:

$$\int \left(\frac{5}{3}\right)^x dx = \frac{\left(\frac{5}{3}\right)^x}{\ln\left(\frac{5}{3}\right)} + C.$$

Step 3: Final expression

Substituting back the simplified result, the integral evaluates to:

$$\int 5^x 3^{-x} dx = \frac{\left(\frac{5}{3}\right)^x}{\ln\left(\frac{5}{3}\right)} + C,$$

where C is the constant of integration.

Final Answer

The solution to the integral is:

$$\int 5^x 3^{-x} dx = \frac{\left(\frac{5}{3}\right)^x}{\ln\left(\frac{5}{3}\right)} + C.$$